

Jam Kabeer Ali Khan

✉ jamkhan@connect.hku.hk | ☎ (+852) 6902 3473 | 🌐 jam-khan.github.io | 📄 jam-khan

RESEARCH INTERESTS

I work on the theory and implementation of programming languages and formal verification, building tools and systems with *foundational* guarantees. My research spans type systems, program logics, compiler correctness, constraint solving, refinement types, and probabilistic cost analysis.

EDUCATION

Sept 2021 – June 2025	University of Hong Kong Major in Computer Science, Minor in Mathematics GPA: 3.5/4.3 Fully-Funded Tuition Scholarship	Hong Kong
Jan 2024 – May 2024	Northeastern University Semester Exchange (Computer Science and Mathematics) GPA: 3.92/4.0 Programming Languages (A), Number Theory (A), Networks & Distributed Systems (A-), Algorithms (A).	Boston, USA
Sept 2018 – July 2020	Nixor College GCE A-levels: 4 A*s & 2 As Regional Distinction in Computer Science, Physics Fully-Funded Scholarship	Karachi, Pakistan

PUBLICATIONS & MANUSCRIPTS

Draft	Internalizing Separate Compilation and Linking via First-Class Environments Jam Kabeer Ali Khan, Jinhao Tan, Bruno C. d. S. Oliveira
Submitted@POPL'27	Foundational Constraint Solving for Expressive Refinement Typing Jam Kabeer Ali Khan, Petros Markopoulos, Nico Lehmann, Ranjit Jhala
Submitted@POPL'27	Polymorphic First-Class Environments Jinhao Tan, Jam Kabeer Ali Khan, Bruno C. d. S. Oliveira
ArXiv	Traq: Estimating the Quantum Cost of Classical Programs Anurudh Pedhuri, Jam Kabeer Ali Khan, Gilles Barthe, Michael Walter.
LICS'26	Complete Relational Logic for Infinite-Dimensional Quantum Programs with Unbounded Assertions Gilles Barthe, Minbo Gao, Jam Kabeer Ali Khan*, Matthijs Muis, Ivan Renison, Keiya Sakabe, Michael Walter, Yingte Xu, Tianshi Yu, Li Zhou. * alphabetical order
PLDI'25 SRC	Capabilities as First-Class Modules with Separate Compilation Jam Kabeer Ali Khan Gold Award (Undergraduate Category)

RESEARCH EXPERIENCE

Jan 2026 – Present	Research Collaboration, UC San Diego Research (<i>Lean4</i> , <i>Rust</i>); Collaboration with Ranjit Jhala and Nico Lehmann	Remote
	- Designed and implemented a Lean 4 framework for constraint solving of refinement types. - Novel results include deterministic proof term synthesis and certifying predicate abstraction. - Mechanization of foundational verifiers to emit CHCs and evaluation of solver on Flux VCs. - Design and implementation of prophecy variables in Flux.	
Apr 2025 – Sept 2025	Max Planck Institute for Security & Privacy Research Intern; Supervised by Gilles Barthe ; collaborated with Michael Walter	Germany
	- Investigated compiler correctness in probabilistic/unitary settings and quantum cost analysis. - Formalized mathematical infra (Dirac notation, Orthonormal Bases) in Lean4. - Designed and implemented extraction from an interactive prover using OCaml to well-typed Lean4 specifications.	

Aug 2024 – Apr 2025; Nov 2025 – present (part-time)	HKU Programming Languages Group Research; Supervised by Bruno C. d. S. Oliveira	Hong Kong
	- Research on the design and implementation of source languages with first-class environments. - Coq mechanization of type-directed elaboration and semantics preservation from source to core calculus. - Prototypes in OCaml and Haskell with property-based testing, along with mechanized proofs in Rocq.	
June 2024 – Aug 2024	HKU Quantum Information and Computation Lab Research Intern; Supervised by Ravishankar Ramanathan	Hong Kong
	Utilized semi-definite programming and sum-of-squares decomposition to design DI-QKD protocols with self-testing properties.	

INDUSTRIAL EXPERIENCE

Oct 2025 – Present	Standard Chartered <i>Functional Programmer (Haskell), Full-time</i>	Hong Kong
	- Contributed to an industrial compiler for Mu, a strict variant of Haskell used for quant finance. - Optimized the Shake build system powering Mu's library infrastructure. - Contributed to a protoc plugin to compile Protobuf to Haskell.	

PRESENTATIONS & TALKS

PLanQC '26	Traq: Estimating the Quantum Cost of Classical Programs [Slides] [Extended Abstract] [Talk]
PLDI '25 SRC	Capabilities as First-Class Modules with Separate Compilation [Talk] [Poster]
HKU Seminar	Maximal Violation of a Measurement Protocol with Simplex Bell Inequality [Slides]

SOFTWARE

Flex: Foundational Constraint Solver <i>Lean4</i> [GitHub] Certified <i>Constraint Horn Clause</i> (CHC) solving to enable expressive refinement typing. EnvML: Polymorphic First-Class Environments <i>Haskell</i> [Playground] [GitHub] An ML-like source language with elaboration to a core calculus, a variant of System F with first-class environments. λ_{SCE}: Mechanization and Implementation <i>OCaml, Lean4</i> A novel calculus with mechanized proofs of type-soundness, semantics preservation, and separate compilation correctness. CQOTL: Relational Prover for Classical-Quantum Programs <i>OCaml, Lean4</i> [Lean4 Mechanization] [GitHub] An interactive prover based on a sound and complete relational logic for classical-quantum programs with backend to Lean4. Traq: Tool to estimate quantum speedups in classical programs <i>Haskell</i> [GitHub] A Haskell package with probabilistic source language, CPL, to perform source-level quantum cost analysis.	
--	--

TEACHING EXPERIENCE

Fall '23, Fall '24	Teaching Assistant , University of Hong Kong COMP2396 Object-Oriented Programming in Java
Spring '23	Teaching Assistant , University of Hong Kong ENGG1340 Computer Programming (C++)

OTHER EXPERIENCE

Apr 2022 – Oct 2022	HKU Sleep Lab <i>Student Research Assistant</i>	Hong Kong
June 2023 – Aug 2023	ActusRayPartners Limited Hedge Fund <i>Technology Intern</i>	Hong Kong

Mar 2023 – Nov 2023	All Round Education Academy <i>Software Developer (Freelance)</i>	Hong Kong
Feb 2022 – Aug 2023	Tutor (Math & CS) , All Round Education Academy	Hong Kong

HONORS & AWARDS

2025	PLDI'25 SRC Gold Award
2022 – 2024	HKU Undergraduate Entrance Scholarship for Outstanding Academic Talents
2021 – 2024	HKSAR Government Scholarship Fund
2024	Rosita King Ho Scholarship
2023	Cathay Hackathon: 2 nd runner-up
2021	HKU Foundation Entrance Scholarship
2020	Top in Sindh and Balochistan (GCE A-levels CS & Physics)
2018–20	100% merit scholarship to attend GCE A-levels at Nixor College.

SKILLS

Languages	Haskell, OCaml, C++, Racket, Java, Python
Proof Assistants	Rocq, Lean4
Tools	Git & L ^A T _E X

REFERENCES

Gilles Barthe

gilles.barthe@mpi-sp.org

Scientific Director

Max Planck Institute for Security and Privacy

Bochum, Germany

 Google Scholar

Ranjit Jhala

rjhala@ucsd.edu

Professor

University of California, San Diego

San Diego, USA

 Google Scholar

Bruno C. d. S. Oliveira

bruno@cs.hku.hk

Associate Professor

University of Hong Kong

Hong Kong

 Google Scholar